

FRONT PANEL • 406 × 305 MM (16 × 12 IN) • ALL DIMENSIONS IN MM FROM THE TOP-LEFT CORNER

Bench Meter Front Panel

Hole layout for the enclosure door or front plate. The EA7-T8C sits top-left. The controls are in a column on the right, and the three jack pairs run along the bottom. The 480 V AC jacks sit in their own marked zone, as far as the panel allows from the DC and Ω jacks.

Layout — front view

305 (12 in)

406 (16 in)

36.25 →

EA7-T8C CUTOUT

235.0 × 179.7

bezel outline 276.7 × 222.2

DC V



+ · COM · 30 V MAX

Ω · CONT



Ω · COM · NO VOLTAGE



POWER
OFF · ON



ON



HOLD



REL
zero



BEEP

AC V · 480 V MAX



L1 · L2 · CAT III 600 V

cutout 35.85 → 270.85

Solid dark shapes are the holes you cut. Dashed outlines are what shows on the front: the HMI bezel and the 30 mm bezels of the 22 mm devices. Keep other holes outside them. The AC jacks are about 200 mm from the Ω jacks, and a probe lead can't plug into the wrong pair by accident. The jacks in each pair are 19.05 mm (0.75 in) apart, so standard dual banana plugs fit.

Hole schedule

ITEM	PART	HOLE	CENTRE X, Y (MM)	LEGEND
DISPLAY				
HMI	EA7-T8C	235.0 × 179.7 rect	153.35, 126.10	Cutout edges: X 35.85–270.85, Y 36.25–215.95. The bezel covers 276.7 × 222.2.
CONTROLS (22 MM, 30 MM BEZELS)				
S1	GCX3300 selector, 2-pos maintained	Ø 22.3	330, 45	POWER · OFF / ON
PL1	ECX2052-24L green LED pilot, 24 V	Ø 22.3	380, 45	ON
PB1	GCX1100 momentary, black, 1 N.O.	Ø 22.3	330, 110	HOLD → CPU X1
PB2	GCX1100 momentary, black, 1 N.O.	Ø 22.3	380, 110	REL → CPU X2
BZ1	ECX2070-24 buzzer	Ø 22.3	355, 175	BEEP → CPU Y1
INPUT JACKS (POMONA 72913)				
J1 / J2	72913-2 red · 72913-0 black	Ø 14.5	40, 275 · 59.05, 275	DC V · + / COM · 30 V MAX
J3 / J4	72913-2 red · 72913-0 black	Ø 14.5	115, 275 · 134.05, 275	Ω · CONT · Ω / COM
480 V ZONE				

ITEM	PART	HOLE	CENTRE X, Y (MM)	LEGEND
J5 / J6	72913-2 red · 72913-0 black	Ø 14.5	336.5, 275 · 355.55, 275	AC V · L1 / L2 · 480 V MAX · CAT III. Red border and warning label.

Wiring behind the panel

- **S1 POWER** switches the 120 VAC line (L) between the fuse and the PSB24. The GCX3300 contact block is rated for mains, but check that it's 1 N.O. and order a second block if you want to switch both L and N.
- **PL1 ON** goes across +24 V / 0 V (TB10 / TB12 on the stripboard). It lights when the 24 V supply is up.
- **PB1, PB2** each go from +24 V (stripboard TB11 "PB") to CPU X1 / X2.
- **BZ1:** (+) to CPU Y1, (–) to 0 V (stripboard TB13 "BUZ–").
- **J1–J4** go to the stripboard through F3 / F4 as before. **J5 / J6** go only to F1 / F2 and the VACT.

Enclosure

- **Front:** any metal enclosure with a 16 × 12 in (406 × 305 mm) or larger face. Bond the door to the enclosure ground with a braided strap.
- **Depth:** at least 6 in (150 mm). The EA7-T8C needs 44 mm behind the panel, plus room for the EA-2CBL connector. The DIN rail parts (CLICK, PSB24, VACT, fuse holders) mount on the back plate.
- **Cord:** bring the 120 VAC cord in through a strain-relief cord grip on a side wall, with the 2 A fuse holder next to it, not on the front.
- **Panel thickness:** check the maximum for the HMI mounting clips and the Pomona jacks before choosing plate thicker than 14 ga (1.9 mm).

Before cutting

1. The drawing is proportional but not full size, since the panel is bigger than a sheet of paper. Lay out the holes from the hole schedule with a square and centre punch, then hold each part up to its mark before drilling.
2. Check the EA7-T8C cutout against the panel's own instructions or datasheet. The figures here are from AutomationDirect's EA7-T8C spec sheet.

3. Behind the panel, keep the 22 mm device bodies (about 60 mm deep) clear of the HMI's rear housing, which is 232.6×177.6 mm, centred on the cutout.